

STEPPERONLINE®

ESS23-RS Series

**Bus Type Integrated Stepper Motor
User Manual**

Version: V1.0

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Foreword

Thank you for using our Bus type stepper motor drive.

Before using this product, be sure to read the manual to learn the necessary safety information, precautions, and operating methods.

Incorrect handling may lead to extremely serious consequences.

Statement

This product is designed and manufactured without the ability to protect personal safety from mechanical system threats. Users are advised to consider safety precautions during use to prevent accidents caused by improper operation or product abnormalities.

Due to product improvements, the contents of this manual are subject to change without notice.

Our company will not be responsible for any modification of the product by the user.

When reading, please pay attention to the following signs in the manual:



Notice: Remind you to pay attention to the main points in the text.



Caution: Indicates that incorrect operation may result in personal injury and equipment damage.

The contents described in this user manual are only applicable to the following models:

Model	Motor length L(mm)
ESS23-RS10	56
ESS23-RS20	80

1 Overview

1.1 Product Description

ESS23-RS series bus-type integrated stepper motor adds bus communication and single-axis controller functions based on digital stepper drives and traditional digital close-loop stepper drives. The bus communication adopts the MODBUS interface.

The current control technology based on load can effectively reduce the heat of the motor and prolong the service life of the motor. The driver's built-in alarm output signal is convenient for the upper computer to monitor and control. The out-of-position alarm function ensures the safe operation of the processing equipment.

1.2 Feature

- A new generation of 32-bit DSP technology, cost-effective, smooth, low noise and low vibration
- Bus-type drive can achieve long-distance reliable control, which effectively solve the problem of pulse loss in interference environment.
- Users can set the current through the bus, subdivision, lock current, control motor start and stop and inquire real-time status of the motor.
- Support position control, speed control and multi-position mode.
- 4 photoelectric isolation programmable high-speed differential input interface, external signal can be used to control the motor start and stop.
- 2 photoelectric isolated programmable output interface, output drive status and control signals.
- Smooth and precise current control, small motor heat.
- Excellent smoothness in low frequency and small microstep.
- Voltage: DC24-50V.
- Over-voltage, under-voltage, over-current protection.

1.3 Applications

Mainly used in wire stripping machines, marking machines, cutting machines, plotters, medical equipment and automation equipment and instruments.

2 Performance Indicators

2.1 Electrical Features

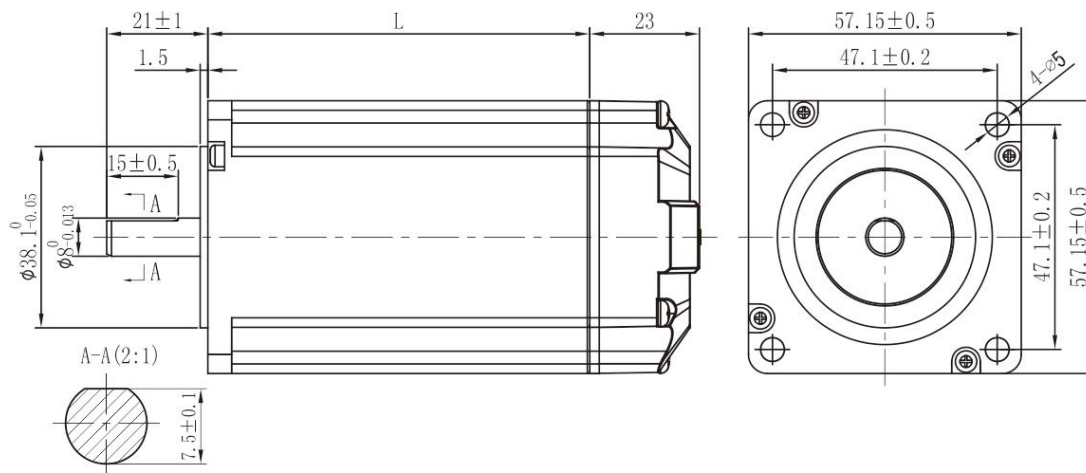
Spec.	ESS23-RS Series			
	Minimum value	Typical value	Maximum value	Unit
Output current	1.0	-	4.0	A
Input voltage	24	36	48	Vdc
Logic input current	10	10	50	mA
Logic input voltage	-	24	24	V
Pulse frequency	0	-	200	kHz
Insulation resistance	100	-	-	MΩ

2.2 Working Environment

Cooling	Cooling fin	
Working environment	Environment	Keep away from other heating equipment as far as possible. Avoid dust, oil mist, corrosive gas, strong vibration, prohibit combustible gas and conductive dust
	Temperature	0℃~50℃
	Humidity	40—90%RH(No dew)
	Vibration	10~55Hz/0.15mm
Storage temperature	-20℃~+80℃	

3 Installation

3.1 Installation Dimensions



Model	The length of motor L(mm)
ESS23-RS10	56
ESS23-RS20	80

Installation dimensions(unit:mm)

3.2 Installation Requirements

The ESS23-RS series bus type integrated stepper motor needs to be installed on a stable base, and the cold air is circulated, which is beneficial to the heat dissipation of the motor. If the installation is not smooth, the internal parts will vibrate and damage.

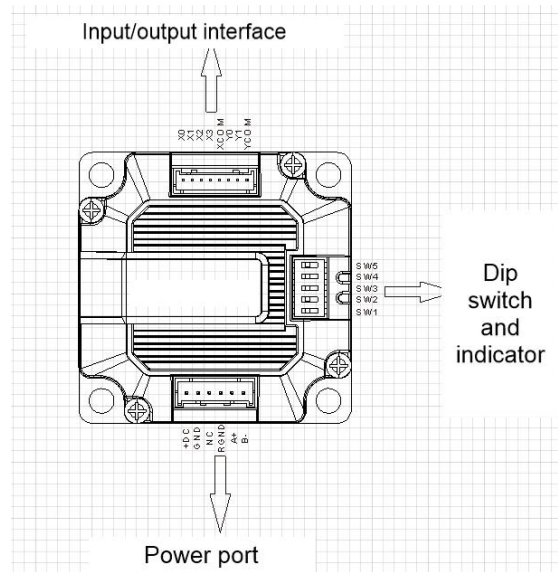
The center axis of the motor rotation is required to be centered and cannot exceed the allowable error range..

The reliable operating temperature of the drive is usually 50℃, the motor operating temperature ≤ 80 ℃.

If necessary, install a fan near the drive to force the heat dissipation to ensure the drive work in a reliable working temperature.

4 Drive Port and Wiring

4.1 Wiring Diagram



Drive side wiring diagram



Caution:

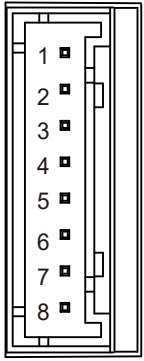
- The personnel involved in the wiring must have professional ability.
- Do not wire with electricity power on.
- Wiring after the installation is firmly finished.
- Do not wrongly connect + and – of power, input voltage should not exceed 50V.

4.2 Port Definition

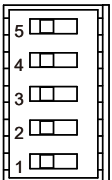
4.2.1 Status light

Color	Symbol	Name	Function
Green	PWR	Power supply indication	When powered on, keep the indicator light on;
Red	ALM	Alarm indication	Current is overcurrent, the indicator flashes once a cycle; In case of overvoltage, the indicator light will blink twice. Under voltage, the indicator light flashes three times; When the error is exceeded, the indicator light blinks five times.

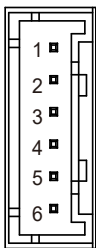
4.2.2 Input/output Port

Port	Pin	Mark	Name	Function
	1	X0	Single-ended input port	Input terminal, signal power supply 24V driver, support NPN and PNP two wiring modes, port function support software modification
	2	X1		
	3	X2		
	4	X3		
	5	XCOM	Single-ended input common	Single-ended input port Public end
	6	Y0	Single-ended output common	The output terminal supports NPN and PNP connection modes, and the port function supports software modification
	7	Y1		
	8	YCOM	Single-ended output common	Single-ended output public end

4.2.3 DIP Switch

Terminal	Pin	Symbol	Name	Function
	1	SW1	DIP switch	SW1: 120 terminal resistor effective bit
	2	SW2		SW2-5: Drive address setting
	3	SW3		
	4	SW4		
	5	SW5		

4.2.4 Power / Communication Terminal

Terminal	Pin	Symbol	Name	Function
	1	B-	Communication port	Modbus communication cable
	2	A+		
	3	RGND		
	4	NC		
	5	GND	Power terminal	DC: 24-50V
	6	+DC		

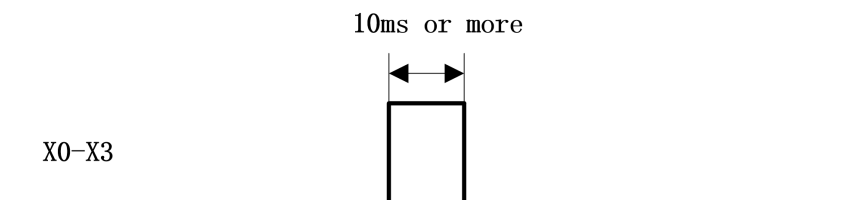
4.3 Input/Output Terminal Operation

● Terminal hardware description

ESS23-RS series drives provide 4 opto-isolated programmable input interfaces, compatible with NPN wiring and PNP wiring.

4 (X0-X3) programmable input signal and external control interface are isolated through optocoupler. The drive is compatible with common cathode and common anode connection, as shown below. In order to ensure that the drive optocoupler conduction is reliable, the controller requires to provide drive current at least 10mA. The drive has been inserted with optocoupler current limiting resistor, Standard input signal voltage is 24V.

The level pulse width of the X0-X3 input needs to be greater than 10ms, otherwise the drive may not respond properly. The X0-X3 timing diagram is shown below:

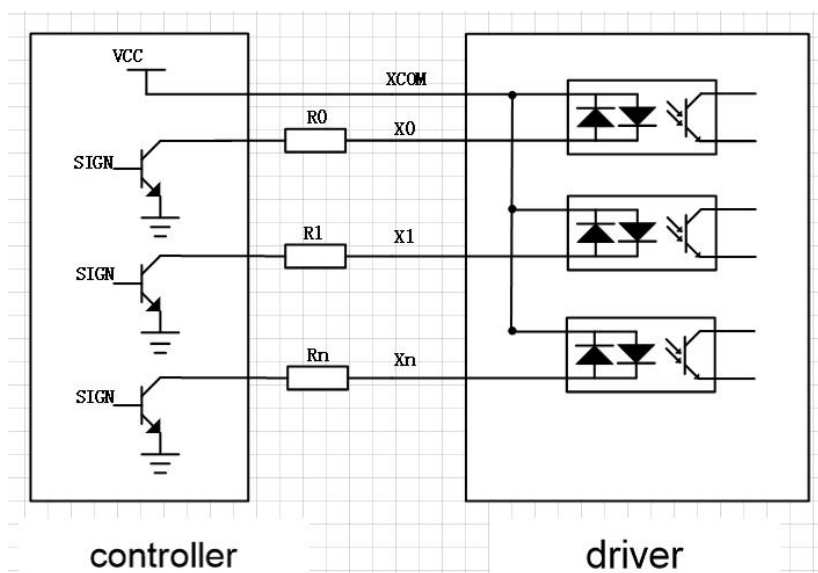


X0-X3 timing diagram

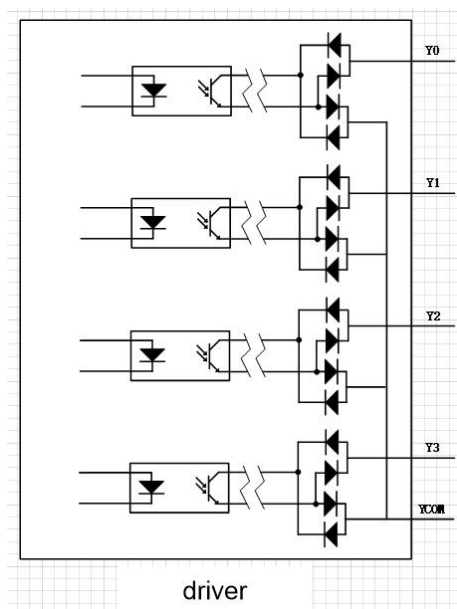
Each time the drive is powered on, X0-X3 defaults to the unspecified state. At this time, the input signal is invalid. The user can configure the X0-X3 input function through the bus.

ESS23-RS series drives provide 2 optocoupler isolated output terminals, support NPN wiring and PNP wiring, and can support high level and low level effective controllers.

● Signal interface wiring diagram



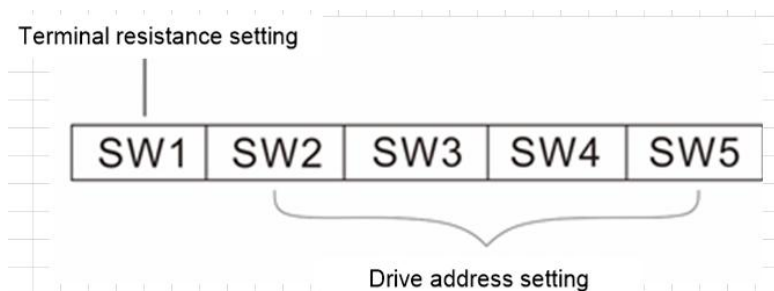
Input signal interface wiring diagram



Output signal electrical schematic

4.4 DIP Switch Setting

The ESS23-RS series bus type stepper motor drives uses a 5-bit DIP switch to set the drive address and terminating resistor. The details are as follows:



DIP switch chart

● Drive Address Setting

The user can use MODBUS bus to control up to 15 ESS23-RS series drives at the same time. The drive communication address setting adopts 4-bit DIP switch, the address setting range is 0~15, where address 0 is Reserved for the system, when the drive address is set greater than 15. It needs to be set and saved using the upper debugging software, and the SW2~SW5 switches must be all set to OFF. As shown in the following table:

SW5	SW4	SW3	SW2	Address
OFF	OFF	OFF	OFF	Customize
OFF	OFF	OFF	ON	1
OFF	OFF	ON	OFF	2
.....				
ON	ON	ON	OFF	14
ON	ON	ON	ON	factory data reset

Note 1: When all 4-bit DIP switches are set to ON, the drive is restored to factory defaults.

Note 2: When all 4-bit dip switches are turned OFF, the driver node address can be customized by the host computer.

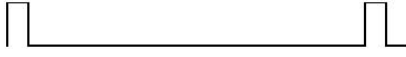
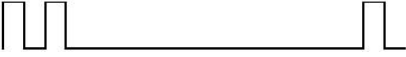
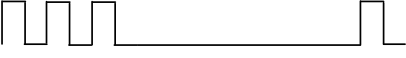
● Terminal Resistance Setting

The user can use this bit to select whether the communication end is incorporated into the 120 terminating resistor. According to the use situation, in general, only the master station and the last slave station need to connect 120Ω of terminating resistor. As shown in the following table:

SW1	120 terminal resistance selection bit
OFF	Invalid
ON	Valid

5 Alarm Diagnosis

ESS23-RS series drive has 4 kinds of alarm information, the alarm indicator flashing several times according to the alarm code, the specific alarm code and treatment as shown in the following table.

Alarm code	Alarm message	Indicator	Reset
Err1: 0x01	Overcurrent or short circuit between phases		Lock motor / re-power to reset
Err2: 0x02	Power supply voltage high		Lock motor / reset automatically
Err3: 0x03	Power supply voltage low		Lock motor / reset automatically
Err5: 0x05	Out of tolerance		Re-power to reset

6 Version Histroy

Version	Description	Time	Remark
V1.0	First edition	2025/04/15	